

Taeseong Kim

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Overview

I am an Integrated Ph.D. student at the Department of Mathematical Sciences, Seoul National University (SNU), Republic of Korea. My advisor is Prof. Jung Hee Cheon. I work on cryptography, currently focusing on Homomorphic Encryption (HE) and its applications, including privacy-preserving machine learning.

Publications

Authors are listed in alphabetical order by last name.

CONFERENCE PUBLICATIONS

Youngjin Bae, Jung Hee Cheon, Minsik Kang, Taeseong Kim[‡]

[C03] HIGH-THROUGHPUT AES TRANSCIPHERING USING CKKS: LESS THAN 1MS

[‡] denotes corresponding author

WAHC

2025

Dongwhan Rho[†], Taeseong Kim[†], Minje Park, Jung Woo Kim, Hyunsik Chae, Jung Hee Cheon, Ernest K. Ryu

[C02] ENCRYPTION-FRIENDLY LLM ARCHITECTURE

[†] denotes co-first author

ICLR

2025

Wonhee Cho, Guillaume Hanrot, Taeseong Kim, Minje Park, Damien Stehlé

[C01] FAST AND ACCURATE HOMOMORPHIC SOFTMAX EVALUATION

ACM CCS

2024

JOURNAL PUBLICATIONS

Jung Hee Cheon, Minsik Kang, Taeseong Kim, Junyoung Jung, Yongdong Yeo

[J01] BATCH INFERENCE ON DEEP CNNs WITH FULLY HOMOMORPHIC ENCRYPTION USING CHANNEL-BY-CHANNEL CONVOLUTIONS

IEEE TDSC

Aug. 2024

MANUSCRIPTS

Jung Hee Cheon, Seungwan Hong, Minsik Kang, Jonghyun Kim, Taeseong Kim, Changmin Lee, Junho Lee

[M01] MRFHE: MIXED-RADIX FULLY HOMOMORPHIC ENCRYPTION WITH BETTER BATCH BOOTSTRAPPING

IACR ePrint 2026/853

2026

Taeseong Kim, Jonghoo Lee, Taeyeong Noh, Jung Hee Cheon, Guillaume Hanrot

[M02] LATENCY-AWARE, HIGH-THROUGHPUT HOMOMORPHIC AES EVALUATION WITH CKKS

IACR ePrint 2026/1209

2026

Talks

2026

PACOH Workshop: Homomorphic Encryption 2026

LATENCY-AWARE, HIGH-THROUGHPUT HOMOMORPHIC AES EVALUATION WITH CKKS

Sokcho, South Korea

June. 2026

2025

Korea Mathematics Society Spring Meeting

HIGH-THROUGHPUT AES TRANSCIPHERING USING CKKS: LESS THAN 1MS

Daejeon, South Korea

Apr. 2025

2024

Korea Mathematics Society Fall Meeting

ENCRYPTION-FRIENDLY LLM ARCHITECTURE

Suwon, South Korea

Oct. 2024

Korea Mathematics Society Spring Meeting

FAST AND ACCURATE HOMOMORPHIC SOFTMAX EVALUATION

Daejeon, South Korea

May 2024

2023

Honors & Awards

AWARDS

- Oct. 2024 **Excellent Prize (3rd Place)**, National Cryptography Contest, NSRI (for [C01])
- Oct. 2023 **Special Prize**, National Cryptography Contest, NSRI (for [J01])
- Dec. 2019 **First Place Scholarship**, Software-Focused University Program (Undergraduate)

HONORS & SCHOLARSHIPS

- 2025–2026 **SNU Teaching and Research Support Scholarship**, Seoul National University
- 2022–2024 **BK 21+ Scholarship**, Ministry of Education of Korea
- 2018–2021 **National Scholarship for Outstanding Students in Science & Engineering**, Korea Student Aid Foundation

Experiences

TEACHING

Seoul National University

Seoul, Republic of Korea

TEACHING ASSISTANT, DEPARTMENT OF MATHEMATICAL SCIENCES

Mar. 2022 - Present

- Introduction to Cryptography
- Calculus for Humanities and Social Sciences 1
- Calculus Practice 2
- Calculus Practice 1
- Differential Equation and Practice
- Linear Algebra

(2025 Winter)
(2024 Winter)
(2021, 2023 Winter)
(2023, 2025, 2026 Spring)
(2022 Winter)
(2022 Spring)

INTERNSHIPS

CryptoLab

South Korea

RESEARCH INTERN (PART-TIME)

Sep. 2025 - Feb. 2026

CryptoLab and FHESLab

France

RESEARCH INTERN (FULL-TIME)

Jul. 2025 - Sep. 2025

T-Brothers

Canada

UNDERGRADUATE INTERN

Jan. 2019 - Mar. 2019

- Developed in-house applications for controlling the flow of stocks.

EDUCATION PROGRAM

University of Southern California (USC)

Los Angeles, USA

SW PIONEER PROGRAM

Jul. 2019 - Aug. 2019

Education

Integrated Ph.D. in Mathematical Sciences

Seoul, Republic of Korea

SEOUL NATIONAL UNIVERSITY

Sep. 2021 - Present

- Advisor: Prof. Jung Hee Cheon (from Jan. 2023).
- Focus: Homomorphic Encryption, Privacy-Preserving Machine Learning.

B.S. in Mathematical Sciences and Industrial Mathematics

Busan, Republic of Korea

PUSAN NATIONAL UNIVERSITY

Mar. 2015 - Aug. 2021

- Summa Cum Laude